



From real estate financialization to decentralization: A comparative review of REITs and blockchain-based tokenization

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ABSTRACT

Blockchain-based tokenization represents a transformative force in the real estate sector, offering a compelling alternative to the well-established model of Real Estate Investment Trusts (REITs). As the real estate industry evolves from financialization to decentralization, driven by technological advancements, these two models offer distinct approaches to democratizing real estate investment. REITs have long been a cornerstone of real estate financialization, allowing individual investors to participate in large-scale real estate ventures through fractional ownership of diversified property portfolios, thus advancing financial supply and share of the housing market. However, the advent of blockchain technology and decentralized finance (DeFi) introduces a new paradigm where real estate ownership can be fractionalized into digital tokens, enhancing liquidity, transparency, and accessibility through global 24/7 trading platforms. The findings suggest that while REITs have successfully broadened the investor base and improved market liquidity, blockchain-based tokenization has the potential to advance these achievements further by lowering entry barriers, reducing transaction costs, and decentralizing market operations. However, the adoption of blockchain technology in real estate also presents challenges, including regulatory uncertainties, technological risks, and the need for robust governance frameworks. As the lines between finance and technology continue to blur, this research underscores the necessity of adapting regulatory frameworks and investment strategies to navigate the evolving landscape of real estate investment. The critical review concludes by discussing the future implications of these trends, highlighting the need for continued research and regulatory innovation to fully realize the potential of decentralized real estate markets, particularly in examining housing inequality and affordability, as housing is not only an important investment vehicle but also as a fundamental shelter for people to live.

1. Introduction

The intersection of technology and finance has consistently fostered innovations in the built environment, particularly evident in real estate investment. Traditionally, real estate is a capital-heavy sector with high entry barriers and transaction costs (Hsieh and Moretti, 2003; Zochling and Phipson, 2011). As the sector grapples with financialization and decentralization, it becomes clear that traditional models are being reshaped by cutting-edge technologies. Real Estate Investment Trusts (REITs), a cornerstone of real estate financialization since the 1960 s, have democratized access to property investments. By allowing smaller investors to participate in large-scale real estate ventures, REITs have opened new avenues for wealth creation and portfolio diversification, liberating property investment from localized businesses to global stock

markets with much higher liquidity and inclusion (Feng et al., 2011; Block, 2012; Brounen and De Koning, 2012).

However, the advent of blockchain technology and decentralized finance (DeFi) heralds a new era in real estate investment characterized by unprecedented liquidity, transparency, security, and accessibility (Schär 2021; Makarov and Schoar, 2022). Blockchain-based tokenization, which converts real estate ownership rights into digital tokens, represents a radical departure from conventional investment methods. This process facilitates fractional ownership and enhances liquidity by enabling 24/7 trading on global blockchain platforms. Furthermore, the potential technological advantages of blockchain open up new possibilities for corporate governance, investment regulation, and market transactions (Moringiello and Odinet, 2023). The implications are profound, potentially democratizing real estate investment further than

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REITs by lowering entry barriers, reducing transaction costs, and decentralizing markets and corporations.

To fully grasp these transformative developments, it is essential to understand the historical trajectory of housing markets, which have evolved from government control and centralization to privatization and commodification, then financialization, and now decentralization. In the post-World War II era, many governments played a central role in housing provision, emphasizing large-scale public housing projects and regulatory oversight. This period of centralization was marked by significant government intervention to ensure housing availability and affordability as a public good (Beauregard, 2001; Hoff, 2008). As economic paradigms shifted in the latter half of the 20th century, there was a marked move toward neoliberal privatization and commodification (Fenton et al., 2013). Housing began to be seen not merely as a necessity but as a commodity and an investment vehicle. Policy changes encouraged private homeownership and real estate development, fostering a market-oriented approach to housing. The subsequent phase of financialization saw housing further entrenched within global financial markets. The advent of mortgage-backed securities and the proliferation of REITs epitomized this trend, transforming housing into a key asset for institutional investors and integrating it deeply into global finance.

Today, we are witnessing the next stage in this evolutionary arc: decentralization. Blockchain technology enables decentralization to revolutionize housing finance by facilitating direct, peer-to-peer, and real-time transactions and reducing reliance on traditional financial intermediaries (Harvey et al., 2021; Schär, 2021). Web 3.0, characterized by decentralization and blockchain technologies, has the potential to reshape the financial instruments for housing development. Blockchain-based platforms, like decentralized autonomous organizations (DAOs), enable shared ownership and management of housing assets, democratizing access to housing and creating alternative models of homeownership. Their embedded smart contracts facilitate automated transactions on a 24/7 basis. Additionally, DeFi protocols allow peer-to-peer lending and borrowing, providing flexible financing options for housing-related purposes. However, these innovations come with risks, such as housing speculation and regulatory challenges, that must be carefully navigated (Aramonte et al., 2021).

This critical review aims to explore and compare REITs with emerging blockchain-based real estate tokenization in the real estate sector. A comparative analysis of REITs and blockchain-based tokenization is particularly pertinent as both models represent innovative approaches to transforming housing equity into liquid, tradable securities. REITs have long served as a vehicle for converting real estate equity into securities, allowing investors to buy fractional shares that own and manage property portfolios. This model has successfully broadened the investor base and improved market liquidity by offering fractional ownership and security-like exchanges (Subrahmanyam, 2007; Highfield et al., 2021). Similarly, blockchain-based tokenization aims to achieve the same goal through a decentralized framework. By converting property ownership into digital tokens, blockchain technology facilitates fractional ownership and enables these tokens to be traded on global platforms, thus enhancing liquidity and further democratizing access (Abdullah et al., 2023; Baum, 2021; Gupta et al., 2020; Zeng et al., 2024). Our objective is to understand this transition as part of a broader movement towards decentralization in real estate, examining how these new models can coexist or potentially synergize with traditional ones. We investigated the structural, operational, and strategic differences between these investment vehicles, providing a comprehensive analysis that highlights their respective advantages and challenges. In this comparative study, we particularly highlight novel technology's systematic impacts on product structure, portfolio governance, and stakeholder relations and its further implications for market practices and reforms.

The significance of this review lies in its potential to inform a wide array of stakeholders, from individual investors and institutional players to policymakers and technologists. By elucidating the comparative

benefits and drawbacks of REITs and blockchain-based tokenization, we aim to provide actionable insights that can guide investment strategies, regulatory frameworks, and technological development. It is informative for housing stakeholders to understand, regulate, and reform the emerging and fast-evolving tokenization practices of the housing sector. Moreover, it reveals the potential, mechanisms, and externalities of technological innovations on complex social developments. In an era where the lines between finance and technology are increasingly blurred, understanding these dynamics is crucial for the housing market, families, individuals, and the broader built environment.

2. The financialization of real estate: REITs as a case study

The financialization of real estate refers to the process by which real estate assets have been increasingly integrated into the global financial system, transforming them from tangible properties into financial instruments that can be traded, securitized, and leveraged. This shift has been driven by the growing importance of capital markets, the expansion of institutional investment, and the evolution of financial products that directly link real estate to financial markets. As a result, housing and commercial properties are no longer viewed merely as physical spaces but as assets capable of generating significant financial returns. This transformation has had profound implications for how real estate is valued, bought, sold, and managed.

REITs are corporations that own, manage, or finance income-producing real estate across various property sectors. These entities allow individual investors to buy shares in commercial real estate portfolios, thereby democratizing access to large-scale, diversified real estate investments (Ambrose et al., 2005). The concept of REITs was legislated in 1960 under the Eisenhower administration, enabling public participation in commercial real estate on a broad scale (Glascok and Lu-Andrews, 2014). REITs have profoundly transformed the landscape of real estate investment, making it accessible to a wider audience and significantly impacting housing sectors globally. This democratization has shifted the focus from direct housing investments to shared ownership through REITs, opening new avenues for investment.

The 1960s marked the nascent stage of REITs, with a notable shift in focus during the 1980s toward mortgage REITs. However, by the end of the decade, these faced challenges due to a rate squeeze. The Tax Equity and Fiscal Responsibility Act of 1986 was a significant turning point, allowing REITs to manage properties directly, thereby adopting a more active management approach and transforming their role in the real estate sector. The 1990s witnessed explosive growth in REITs, with the sector expanding from \$8 billion to \$50 billion and significant public offerings that marked the beginning of the modern REIT era (Feng et al., 2011). This period underscored REITs' role in enhancing real estate market liquidity, making it more resilient to economic fluctuations. The introduction of innovative structures like UPREITs in the early 1990s further modernized the market by allowing property contributions in exchange for operating partnership units, enhancing flexibility in property management and investment.

REITs' resilience was tested during the market downturns of 1998–1999 and the global financial crisis of 2007–2008. Despite these challenges, REITs managed to sustain and grow, partly thanks to their structure, which mandates significant payout ratios to maintain investor confidence. The inclusion of REITs in the S&P 500 Index in 2001 underscored their importance in the financial landscape, marking them as a viable and stable investment route. REITs have not only made investment opportunities more accessible in the housing sector but have also played a pivotal role in shaping the modern real estate market. Their ability to adapt to economic changes and integrate new technologies highlights their resilience and enduring relevance in the market.

3. The rise of tokenization in real estate via blockchain

Tokenization of housing properties refers to the mechanism of

issuing digital tokens on a blockchain that represent the entire or fractional ownership of a physical real estate asset. Bought, sold, and traded on digital exchanges, the token enables investors to invest in smaller portions of traditionally capital-heavy real estate assets, lowering the barrier for real estate investment. It emerged along with the development of blockchain technology and encrypted products. As a decentralized ledger instrument, blockchain technology advances transparency, security, and immutability of transactions, making it favorable for transitions of valued assets (Simons and Simons, 2022). Based on the projected assets, tokens could be categorized into digital currencies (e.g., Bitcoins), utility tokens (used solely in a network), and security tokens (representing investment contracts like securities, properties, and commodities).

Tokenization represents the financial fractionalization and transactional digitalization of real estate properties. As discussed above, real estate investment has already gone through processes of financialization and fractionalization with products like REITs, timeshares, and syndication. The revolutionary advantage of tokenization is rooted in blockchain technology and derived digital exchanges. With Ethereum-based smart contracts that can self-execute contracts with terms written into code, blockchain technology facilitates the automated transactions of token transactions, rent collections, and profit distributions without agents or intermediaries (Fang et al., 2022a,b; Ullah and Al-Turjman, 2023; Norta et al., 2018). Eliminating intermediaries could substantially reduce transaction costs, making real estate transactions more efficient, inclusive, and affordable (Landy and Wilka, 2018; Zheng et al., 2020). Furthermore, the digital platforms that trade these property tokens enable investors to participate in real estate investment with smaller capital, higher liquidity, faster and safer transactions, and lower costs, expanding the inclusivity of the sector and further digitalizing, decentralizing, and democratizing real estate investments (Danielsen et al., 2014; Gupta et al., 2020; Ke and Yin, 2024).

The operational mechanism of housing tokenization follows a roadmap centered around token offering and trading. The first step is locating the underlying assets with property selection, acquisition, assessment, and possible bundling. The second step is establishing the ownership structure and legal framework in compliance with regulatory requirements. The third step is creating digital tokens by dividing the property(s) into fractional ownership units and coding the terms into smart contracts. The last step is listing digital tokens to the public through initial coin offerings (ICO) and possible digital exchanges. After ICOs, tokens could be traded on secondary markets with higher liquidity than traditional real estate properties. Through smart contracts and DAOs, token holders, or fractional property owners, could actively participate in the return distribution, capital appreciation, and governance of real estate properties (Li et al., 2019; Serrano, 2022).

The development of tokenized real estate practices is still an evolving story. Initial significant practices of real estate tokenization emerged around 2017 when initial coin offerings (ICO) and digital exchanges boomed. Early practices, represented by platforms like Propy and Atlant, focused on the feasibility of fractionalizing real estate ownership in response to the market illiquidity and high entry barriers in traditional real estate markets (Konashevych 2020; Baum, 2021; Jia et al., 2024). Later, between 2019 and 2020, the maturity of blockchain technology, particularly smart contracts, cultivated the concept of Title Tokens, in which a digital record and legal framework of ownership is stored on blockchain. Starting in 2020, pioneering real offerings of real estate tokens were launched on platforms worldwide like BrickX (Australia), Kasa (South Korea), and ADDX (Singapore), demonstrating the benefits like elevated liquidity, inclusive investors, and lower transaction costs while posing novel challenges to regulatory authorities (Chow and Tan, 2022).

Normative institutional investors are gradually realizing and leveraging tokenization's unique features in real estate financing, transaction, and portfolio management. Recent research reveals that tokenized assets show distinct return-risk profiles from traditional assets

like stocks. Their correlation with macroeconomic and debt factors make them attractive for portfolio diversification (Steininger, 2023). A move from REITs to real estate tokens is emerging as several assets like the Aspen St. Regis Resort, despite already cleared SEC (Securities and Exchange Commission) pre-filing processes, shifted from public security offerings (Initial Public Offering, IPO) to digital token issuance (Initial Coin Offering, ICO). The regulatory environment is also embracing tokenized real estate with clearer guidelines, like Germany's Electronic Securities Act of 2021, tailoring investment tunnels for professional investors and demonstrating regulatory support for institutional engagement (Henker et al., 2023).

By leveraging blockchain technology in a digital era, tokenization showcases the potential to revolutionize traditional housing markets regarding investment, transaction, and management. It democratizes real estate investments with multi-dimensional benefits while facing challenges like due diligence flaws, regulatory uncertainties, and lack of standardized protocols and market acceptance. Therefore, it is crucial to scrutinize this emerging technology through comparative reviews with normative paradigms like REITs.

4. Comparative analysis: from financialization to decentralization

The comparison between REITs and blockchain-based tokenization is structured in three aspects: organization, market, and governance. Table 1 below provides a comparative analysis of REITs and blockchain-based real estate tokenization, highlighting their differences in structure, governance, liquidity, market accessibility, and technological adoption, as well as their respective advantages and challenges. This comparison underscores the transition from financialization to decentralization in the real estate sector.

4.1. Organizational basis: structure, agencies, and centrality

REITs are structured as centralized entities that manage diversified property portfolios. Investors own shares in the REIT itself rather than in individual properties, benefiting from professional management and diversification (Feng et al., 2011). This model ensures stable income streams through high dividends and facilitates investor trust by adhering to established governance frameworks regulated by entities like the SEC (Ambrose et al., 2005). However, it limits direct ownership, as shares are proxies for real estate assets.

In contrast, blockchain tokenization directly fractionalizes real estate assets into digital tokens, which represent ownership of individual properties. The governance of tokenized real estate assets often includes DAOs, where token holders can participate directly in the decision-making processes of specific assets rather than the portfolio holding entity. This decentralization provides investors with autonomy over their investments, reducing reliance on intermediaries and offering greater control over asset-specific risks. The market has observed personal asset owners tokenizing their properties on various platforms. This eliminates many intermediaries like brokers and asset management companies, empowering personal investors to sell, purchase, and trade fractional ownership of individual properties. Platforms like Propy illustrate this model by enabling investors to own and trade tokenized property fractions on blockchain platforms, bypassing traditional ownership and market barriers (Simons and Simons, 2022).

However, we must recognize that the current adoption of tokenization in established markets is still at a primitive stage. Tokens based on individual properties and owners operate in a decentralized way, but not for institutional investors. Real estate tokens that could attract institutional investors are mainly organized as REITs with a centralized managing entity but traded as tokens on blockchain platforms. This reflects a transitional and compromised mix of REITs and tokens while the markets adopt and pivot.

Table 1
Comparison Between REITs and Blockchain-Based Real Estate Tokenization.

Dimension	REITs	Blockchain-Based Real Estate Tokenization
Structure	Shares in a company with diversified real estate portfolios.	Digital tokens representing fractional property interests.
Liquidity	High liquidity, traded on stock exchanges; subject to market hours and fees.	Potential for higher liquidity via 24/7 trading on blockchain platforms.
Governance and Control	Governed by a board and management team; investor influence is proportional to shareholding.	Innovative governance structures such as DAOs with direct investor participation.
Regulation	Public listing is subject to SEC protocols and investment laws.	Without SEC filing and vetting protocols; still pending legislation.
Market Access	Accessible through stock markets; lower entry barriers.	Global investment with fewer regulatory barriers; inclusive.
Portfolio Structure	A composite portfolio with diverse real estate properties.	Usually focus on a single asset in current practices.
Income Stability	Stable income from regular dividends.	Variable income stability; high potential returns and risks.
Capital Appreciation	Potential for property value appreciation and portfolio growth.	Capital appreciation from property values and token demand.
Transparency and Data Availability	Regulated transparency with periodic disclosures.	High transparency; real-time data on the public ledger.
Risk Profile	Risks moderated by diversification and professional management.	High risk from nascent technology and evolving regulations.
Technological Adoption	Increasing technology adoption, but slower in cutting-edge tech.	Built on blockchain for efficiency, security, and innovation.
Market Maturity	Mature market with established metrics and a broad investor base.	Developmental phase; evolving models and expanding investor base.
Entry and Exit Costs	Lower entry and exit costs; brokerage fees may apply.	Further reduced transaction costs; variable token creation and trading costs.
Geographical Diversification	Geographical diversification through varied property locations.	Granular diversification; international investments with fewer barriers.
Investment Granularity	Larger, aggregated investments; limited control over specific assets.	Fractional investments in specific properties; more control.
Legal and Regulatory Considerations	Well-established legal and regulatory frameworks.	Evolving regulatory landscapes; varied jurisdictional approaches.
Operational Efficiency	Economies of scale; higher overhead costs from traditional processes.	Efficiency through smart contracts and automation; lower overhead.

4.2. Market access, liquidity, and transaction costs

REITs have historically democratized real estate investment, enabling individual investors to access large-scale projects without requiring outright ownership. Compared to traditional real estate markets, REITs allow investors to diversify portfolios, enjoy higher dividends, and mitigate risks without full asset ownership (Wang et al., 2017). Moreover, the inclusion of REITs in stock markets and major indexes enhances real estate liquidity and market correlation, demonstrating the broader financial benefits of real estate financialization (Subrahmanyam, 2007; Cheung et al., 2018). However, REITs face some issuance and registration costs, as well as operational inefficiencies due to reliance on centralized systems and extensive regulatory requirements (Hartzell et al., 2005; Highfield et al., 2021). In contrast, blockchain tokenization lowers these barriers by enabling fractional ownership via online platforms. This approach fosters inclusivity, allowing small-scale investors to participate with minimal capital, aligning with the decentralized finance ethos to address traditional real

estate's inclusion challenges (Gupta et al., 2020).

Liquidity presents another key distinction. REITs offer moderate liquidity through stock market listings, but trading is limited to specific hours, and transaction costs such as brokerage fees persist (Brounen and Koning, 2012). Blockchain tokenization, on the other hand, enables fractional ownership with 24/7 trading on global blockchain platforms, facilitating instant transactions. However, this accessibility introduces volatility, where low trading activity led to price instability, revealing vulnerabilities in emerging blockchain markets.

Operational efficiency further differentiates these models. While REITs benefit from economies of scale, they incur higher costs due to manual processes and reliance on intermediaries (Feng et al., 2011). Blockchain platforms leverage Ethereum-based smart contracts to automate rent distribution, compliance, and transaction settlements, reducing costs and errors (Zheng et al., 2020). The decentralized nature of blockchain allows tokens to be traded globally with minimal capital barriers. Smart contracts streamline transactions by eliminating intermediary costs, encouraging small investors, and increasing market liquidity. Platforms like Harbor illustrate this efficiency, reporting transaction costs 30–50 % lower than traditional methods (Veuger, 2019). Furthermore, the transparent and immutable nature of blockchain records ensures real-time and bilateral verification, enhancing efficiency and transaction transparency (Baptista et al., 2023).

4.3. Risk, regulation, and governance

Governance and risk management are central to distinguishing tokenization from other real estate models. While REITs rely on centralized and robust regulatory oversight, tokenization presents innovative decentralized mechanisms with both opportunities and challenges for investors.

REITs have a long-established governance framework rooted in centralized oversight. Board directors and professional management teams are responsible for ensuring operational efficiency, compliance, and transparency. The legal requirement for REITs to distribute at least 90 % of taxable income as dividends aligns investor and management interests, reducing agency costs (Danielsen et al., 2014). However, this model often limits direct investor influence, as decision-making power is concentrated at the top (Ambrose et al., 2005). In contrast, blockchain-based tokenization uses decentralized governance models, such as DAOs. DAOs empower token holders to vote on property-related decisions, democratizing governance and fostering transparency (Harvey et al., 2021). This model mitigates the information asymmetry common in traditional real estate investments by enabling real-time access to financial and operational data on public ledgers (Zheng et al., 2020). However, DAOs face significant implementation challenges. While inclusivity is a core principle, achieving meaningful participation from a diverse investor base can be difficult, especially when stakeholders lack expertise in real estate management (Konashevych, 2020). Moreover, the absence of centralized oversight introduces risks of misaligned incentives, where uninformed or opportunistic voting decisions could undermine property value and management effectiveness.

The risk landscapes for REITs and tokenization are shaped by their governance structures and market maturity. REITs are relatively low-risk instruments due to their established regulatory frameworks and consistent dividend payouts, which attract institutional investors seeking stability (Cotter and Roll, 2009). Their centralized governance ensures rigorous oversight, but this comes at the cost of slower decision-making and higher operational expenses. In contrast, tokenized real estate faces multifaceted risks due to its nascent stage and reliance on blockchain technology. Key risks include: (a). *Regulatory Ambiguities*: The decentralized nature of tokenization often conflicts with traditional regulatory systems. Platforms like Harbor have encountered significant delays due to legal uncertainties around token offerings and anti-money laundering (AML) compliance (Chow and Tan, 2022; Yu et al., 2024). A lack of standardized global frameworks exacerbates these challenges,

discouraging institutional adoption (Moringiello and Odinet, 2023). (b). *Cybersecurity Threats*: While highly secure in theory, Blockchain systems are not immune to hacking and fraud. Smart contracts, which automate governance and financial processes, introduce vulnerabilities when coding errors or malicious attacks occur. The DAOs hack of 2016, which resulted in a loss of \$60 million in Ether, remains a cautionary tale for decentralized platforms (Zheng et al., 2020). (c). *Market Volatility and Liquidity Risks*: Tokenized markets are prone to higher volatility due to their small investor base and speculative activity. For example, Atlant experienced significant price fluctuations during periods of low trading activity, underscoring the liquidity challenges in emerging tokenized real estate markets. Such volatility can erode investor confidence and hinder broader adoption. (d). *Operational Risks in DAOs*: The decentralized governance model, while innovative, lacks the centralized accountability of REITs. Decision-making inefficiencies in DAOs, such as prolonged voting periods or lack of quorum, can delay critical actions like rent adjustments or property maintenance (Harvey et al., 2021).

While tokenization introduces unique risks, it also offers opportunities to reimagine governance and risk management in real estate. By decentralizing decision-making and enhancing transparency, blockchain technology addresses many inefficiencies inherent in traditional models. For instance, the immutable nature of blockchain records ensures real-time data availability, reducing the information asymmetry that plagues REITs' periodic reporting systems. However, critics argue that tokenization's success hinges on the resolution of regulatory and technological barriers. Without clear legal standards, platforms risk alienating institutional investors who are vital for scaling tokenized markets. Additionally, while blockchain systems can enhance operational efficiency, the high technical expertise required to navigate these systems may deter retail investors, limiting inclusivity. A potential middle ground lies in hybrid models that integrate blockchain's transparency with REITs' centralized accountability. For example, an integration of rigorous regulation in REITs (e.g., high dividend distribution and outside oversight) and decentralized structure of tokenization (e.g., smart contracts and DAO) could be a feasible approach to offer the best of both worlds.

5. Future implications and conclusion

The real estate sector is undergoing a significant transformation from traditional financialization to a decentralized model driven by blockchain technology. This shift represents not merely a technological upgrade but a fundamental change in the ownership, management, and trading of real estate assets. Traditional models, such as REITs, have democratized real estate investment, broadening access for a diverse range of investors. However, blockchain-based tokenization is advancing this democratization further by enabling fractional ownership, increasing liquidity, and reducing transaction costs. As blockchain technology progresses, there is an urgent need for more academic research to comprehend its full implications. Future studies should address the regulatory challenges, technological advancements, and market dynamics of blockchain-based real estate tokenization. Comparative analyses of traditional REITs and tokenized real estate can offer valuable insights into the respective benefits and limitations of each model, aiding investors and policymakers in making informed decisions.

Blockchain-based tokenization introduces a transformative approach to real estate investment. Despite its groundbreaking advantages, tokenization faces several challenges that hinder its broader adoption. The fragmented and inconsistent regulatory environment creates significant uncertainty for both issuers and investors, particularly in cross-border transactions. This lack of clarity undermines institutional adoption and erodes market trust. Furthermore, blockchain technology presents vulnerabilities such as cybersecurity threats, smart contract errors, and the potential for technological obsolescence. Addressing these challenges requires comprehensive implementation of robust safeguards to

ensure market stability and trust. Additionally, tokenization remains in its early stages, characterized by low adoption rates and minimal institutional involvement. Overcoming scalability challenges and achieving mainstream acceptance are critical to the long-term success of tokenized real estate models.

The future of real estate investment is likely to be shaped by several key trends. The increased adoption of blockchain technology is anticipated as it becomes more mainstream, leading to more platforms offering tokenized real estate investments and greater market liquidity. Governments and regulatory bodies will need to adapt to the growing presence of blockchain in real estate, potentially creating new frameworks to address the unique challenges posed by decentralized finance, such as ensuring investor protection and preventing fraud. Additionally, future technological developments, such as improved blockchain protocols and enhanced smart contracts, will further streamline the tokenization process and increase its security and efficiency. The decentralized nature of blockchain presents unique regulatory challenges (Chod et al., 2022). To foster the growth of blockchain in real estate, regulatory bodies need to strike a balance between encouraging innovation and protecting investors. Establishing clear legal frameworks, implementing robust oversight mechanisms, and fostering international cooperation among regulatory bodies will be crucial in creating a harmonized regulatory environment (Cheung et al., 2015). Future technological advancements are likely to have a significant impact on both REITs and tokenized real estate. Enhancements in smart contract technology will enable more complex and secure transactions, reducing the need for intermediaries and lowering costs. Developing interoperability between different blockchain platforms will facilitate seamless transactions and increase the attractiveness of tokenized real estate as an investment option. Advances in cybersecurity will be essential to protect the integrity of blockchain-based transactions and safeguard against threats such as hacking and fraud.

The comparative analysis between REITs and blockchain-based tokenization highlights several key points. Both models democratize access to real estate investment, but tokenization advances this further by enabling fractional ownership. Tokenization offers potentially higher liquidity through 24/7 trading on blockchain platforms. Additionally, blockchain provides increased transparency with real-time data access on a public ledger. While REITs operate within well-established regulatory frameworks, tokenized real estate faces evolving regulatory landscapes. Furthermore, tokenization enhances operational efficiency through smart contracts and automation, reducing overhead costs (Gupta et al., 2020). Meanwhile, we observed a distinction that decentralized structures were better implemented in the tokenization of individual properties while institutional investors are more into mixed structured assets with REITs-style organization and blockchain-based token transactions. This suggests that decentralized real estate finance through tokenization is still an evolving story that requires diverse stakeholders' further adoption, transformation, and innovation.

Based on these findings, several recommendations can be made for different stakeholders. Investors should consider diversifying their portfolios by including both REITs and tokenized real estate to balance the benefits of traditional and innovative investment models. Policy-makers should develop clear regulatory frameworks that support the growth of blockchain in real estate while ensuring investor protection. Developers should leverage advancements in blockchain technology to create more efficient and transparent real estate investment platforms. Emerging technologies, like crowdsourcing, data mining, and AI systems, could potentially enhance the performance of market efficiency and investment portfolios (Fang et al., 2022; Wu, 2024). At the same time, stakeholders and scholars must critically examine the potential challenges associated with tokenization. While increased liquidity and market efficiency are touted as advantages, they may inadvertently amplify socioeconomic inequalities by encouraging speculative behavior rather than addressing structural issues like housing affordability. Without thoughtfully crafted policies and interventions,

tokenization risks fostering profit-driven mechanisms that could deepen existing disparities in the real estate market, ultimately hindering its potential to support more inclusive and equitable urban development.

In conclusion, the progression of real estate investment from financialization to decentralization signifies a profound transformation in market dynamics. While traditional REITs have facilitated broader access to real estate assets, blockchain-based tokenization introduces the potential for enhanced inclusivity, transparency, and operational efficiency. As technological advancements continue and regulatory frameworks adapt, the real estate sector is positioned for significant innovation that could benefit investors, developers, and broader market stakeholders. However, this evolution necessitates a cautious approach, underpinned by the development of robust regulatory structures and equitable policies. Given that housing extends beyond its role as an investment vehicle to serve as a fundamental human need, it is imperative to ensure that these innovations align with the principles of social equity and contribute meaningfully to improving living conditions.

CRedit authorship contribution statement

Ang Liu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Cheng Chen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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The two authors contributed equally to this manuscript and are designated as co-first authors.

Data availability

No data was used for the research described in the article.

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